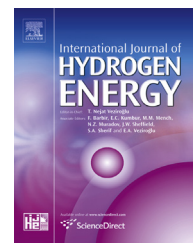


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Full Length Article

Investigation of technical and economic aspects for methanol production through CO₂ hydrogenationKonstantinos Atsonios^{a,b,*}, Kyriakos D. Panopoulos^a,
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ABSTRACT

This study investigates various design and operating aspects for the valorization of industrially captured CO₂ towards methanol production and the circumstances under of which this concept can be economically viable. Cost breakdown in various Power-to-Fuel concepts confirm that hydrogen cost is the most crucial factor. Several power delivery options for hydrogen production through electrolysis are compared for their economics. The use of cheap electricity in conjunction with adequate time coverage throughout the annum is of high importance for lowering the overall H₂ production costs. Also, a Power-to-Fuel system integrated with a coal fired power plant can be an interesting option for excess power transformation when the electricity sell is not profitable. The economic analysis on the H₂ production scheme revealed that each of the three main parameters for the determination of the H₂ cost (electrolyzer capital cost, electricity cost and storage cost) can play the key role in the feasibility of the plant depending on the concept each time. A considerable effort is needed in order the CO₂ derived fuels to reach a competitive level in the global market.

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Introduction

The control of the Greenhouse Gas (GHG) emissions is one of the most challenging environmental issues that should be faced in the 21st century. The increase of CO₂ concentration from 280 ppm at the beginning of the Industrial Revolution to

370 ppm up-to-date is proved that a great part of it comes from anthropogenic factors and has disastrous effects on the global weather, the climate, and the average temperature [1,2]. According to IEA Blue Map Scenario for reducing CO₂ emissions, the Carbon Capture and Storage (CCS) is considered among the major measurements that should be addressed in large scale worldwide [1,3]. However, several

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obstacles such as high capital and operational cost, the several social and technical issues that should be addressed for the CO₂ safe storage in conjunction to the low carbon pricing policy, hinder the adoption of this option for CO₂ footprint mitigation.

A more feasible option to mitigate CO₂ emissions is to transform it into valuable compounds, like fuel organic and inorganic chemicals, namely as ‘the CO₂ Capture and Utilization (CCU) concept’. The majority of CO₂ use in industry is for urea production, which accounts for more than half of the global annual usage [4]. Alternatively, CO₂ is utilized also physically in various applications such as refrigerant medium, in fire extinguishers and in the petroleum and NG industry for Enhanced Oil Recovery (EOR) and Enhanced Gas Recovery (EGR), respectively [3,5]. Even though CO₂ is a thermodynamically and kinetically stable molecule, various ways have been developed for CO₂ reduction, mainly based on the fact that the central carbon of the CO₂ molecule is electrophilic and can be easily attacked by nucleophiles [6]. The methods for CO₂ transformation can be sorted in six categories [4,7]: chemical reduction (i.e. Boudouard), electrochemical reduction [8], photochemical reduction (i.e. artificial photosynthesis), thermochemical conversion (i.e. dry reforming and hydrogenation), biological (i.e. photosynthesis, anaerobic conversion) [9] and inorganic transformation. This study focuses on synthesis of methanol from CO₂ through catalytic hydrogenation. Hence, the main feedstock apart from CO₂ that is required is pure hydrogen.

Since the H₂/CO₂ ratio for CO₂ hydrogenation towards hydrocarbons synthesis should be four (4) for methane and three (3) for methanol synthesis, the required amounts of hydrogen are very large. There are three routes for non-fossil derived hydrogen production: water electrolysis, biomass conversion and solar conversion [10,11]. Electrolysis is based on the water splitting into H₂ and O₂, the energy for this reaction is given in the form of electricity derived from Renewable Energy Source (RES) i.e. photovoltaic panels (PV), wind farms hydropower and geothermal plants. The second category consists of thermochemical (i.e. gasification, pyrolysis) or biochemical conversion of biomass, the intermediate products of which process are properly converted into rich H₂ fuel. Through solar conversion, either via thermolysis or photolysis, heat or photons are respectively used for hydrogen synthesis [12,13]. Biomass gasification and separation of H₂ from the produced product gas is not commercially available at large scales while thermolysis and photolysis processes are just beginning to be explored for their potential use by research projects in the EU [14,15]. In the present study, the hydrogen is considered to be derived from electrolysis, since this is the most mature and well-established technology even in industrial scale [16] and also it is not relied on carbon-contained source like biomass.

Several electricity sources have been proposed in the literature, including cheap grid electricity [17], surplus electricity from power plants [18,19], renewable electricity [20,21], etc. In the case of RES one of the major issues that should be addressed is the stochastic nature of the renewable power production and the extreme load fluctuations that require a highly flexible system for an efficient operation. Besides, the CCU unit cannot operate with maintain high performances

under these variable throughputs conditions as it incorporates, according to the existing technology, since components such unit operations that have lower efficiencies at part load, such as compressors, steam production cycle and the CO₂ capture system itself. A technical solution to this issue is the intermediate storage of the produced H₂ in order to secure the constant hydrogen delivery to the CCU unit with the negative effect of the increase in the investment cost. Methanol is considered as one of the most valuable chemicals with a series of uses in various sectors (power, transport, steel, chemical industry) either as fuel or as block for the synthesis of other chemicals (dimethyl ether, formaldehyde, methyl tert-butyl ether, acetic acid, gasoline, etc). Moreover, since methanol has the lowest production cost to market price ratio among the fuels that can be potentially produced from CO₂, it is the most representative product of the so called “Power-to-Fuel” concept. In Iceland, a demonstration plant with capacity 4000 tons of renewable methanol per year is operating since 2011 and a larger plant with capacity 40,000 t/y is planned to be constructed by Carbon Recycling International (CRI) [22]. In Canada, Blue Fuel Energy will built a methanol plant with capacity 400,000 t/y of methanol, powered by renewable electricity [23].

The scope of this study is the investigation of various aspects for the valorization of CO₂ towards the production of liquid fuel (methanol), in large scale applications. These aspects are related to the power source for the hydrogen production through electrolysis (grid electricity, renewable energy source), the CO₂ capture scheme (selection of the proper capture technique) and the input gas (H₂ and CO₂) transportation options (pipeline, truck or trailer). The assessment is made in terms of minimization of methanol production cost by performing the economic evaluation, determining the boundary conditions and the next steps that should be done so that the proposed concept can be profitable.

Process description

The concept for liquid fuel (methanol) synthesis through CO₂ hydrogenation is shown in Fig. 1. The pure CO₂ stream is fossil derived and comes from separation and purification from combustion gases of a power plant or other intensive carbon emission industry (e.g. cement plant, steel industry).

The concept that is presented in this study is not based on a total utilization of the produced CO₂ from fossil fuel combustion (power plant or intensive industry). In case that the total amount of the CO₂ captured from the flue gases of a typical 300 MW_e coal fired power plant undergoes to hydrogenation, the required power duty for water electrolysis is estimated at 1.77GW_e which is technically infeasible and economically unprofitable.

Several studies dedicated to the techno-economic investigation of this concept can be found in the literature [17,19,24–29]. Among the major conclusions that are extracted from them is that the hydrogen production cost is most significant factor that affects the economics of this methanol-from-CO₂ concept.

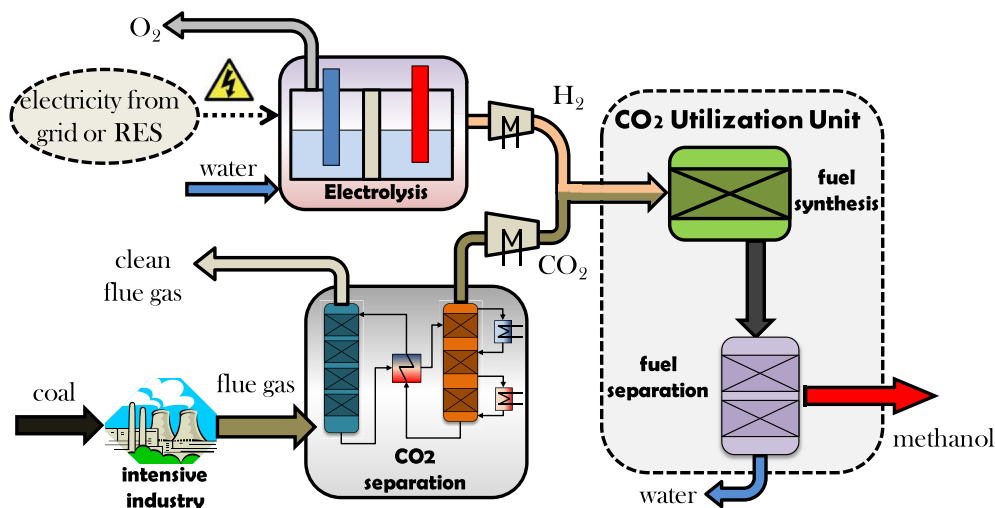
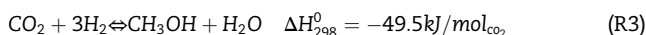
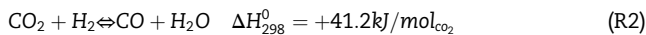
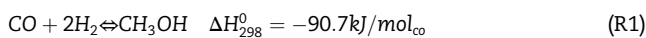


Fig. 1 – Conceptual design of the proposed methanol production from CO₂.

Methanol synthesis from CO₂

Concerning the CO₂ fixation step towards the production of valuable chemicals, the catalytic hydrogenation for methanol is examined. Methanol is formed from CO₂ hydrogenation under three-phase conditions [30]:



Methanol is produced via the exothermic reactions (R 1) and (R 3) while the reverse water-gas-shift (rWGS) reaction (R 2), which is strongly endothermic, occurs in parallel. The adopted methodology for the methanol synthesis is a kinetic model that is presented by Graaf et al. [31].

The process flow diagram of a typical methanol synthesis unit from CO₂ is shown in Fig. 2. The unit consists of the methanol synthesis, gas separation and product purification. The inlet gas is heated up to a temperature level so that the reactor operates adiabatically under the specified temperature. The required heat for crude methanol heating and separation is obtained from the gas outlet cooling. In order to achieve high purity levels in the methanol product (>99.2%), the liquid stream after the first flash separator is throttled down to some kPa below the atmospheric pressure, following the same approach with the Rectisol™ process. A small portion (<1%) of the recycling gas is extracted as purge gas in order to avoid by-products accumulation such as hydrocarbons, inert gas etc.

Table 1 presents the main specifications of the process modeling design of the methanol production unit. This process model is served for the calculation of the scaling parameters that are required for the total equipment cost estimation (see Table 3). The process modeling is necessary at this stage in order to make the energy balance calculations and to determine parameters such as the rejected heat duty

and the power consumptions in the compressors and the blowers, which are not easily calculated otherwise. For simplicity reasons these operating parameters are constant regardless of the plant size. The methodology for the development of the process design for the methanol synthesis and purification is inspired mainly from Ref. [24]. The chosen reactor pressure (65 bar) and heat exchangers (preheaters and gas coolers) and flash separators temperatures allow the methanol production and recovery without any addition of external heat. The methanol synthesis reactor was approached as plug flow reactor with kinetic parameters from the literature [31]. Table 2 presents the main characteristics of the basic streams and their specific to methanol production molar flow rates (i.e. kmol per kmol of produced methanol). The CO conversion under the specified operating conditions is 30.5% while the corresponding conversion when chemical equilibrium occurs is 35.3%. Assuming the pure hydrogen streams is delivered from the electrolysis unit at 30 bar, the specific power consumption for H₂ compression is 78.9 kWh/t_{MeOH}. Furthermore additional 34.5 kWh/t_{MeOH} are required recompression of the recycling gas and pumping the crude methanol. Hence, the total specific power consumption of the methanol plant totals 113.4 kWh/t_{MeOH} (which are taken into account as a cost element). The total heat demands for crude methanol preheating and the distillation column

Table 1 – Key Process specifications for the conventional MeOH synthesis from CO₂.

H ₂ /CO ₂ ratio	3.0
Inlet gas preheating temperature	150.7 °C
MeOH reactor temperature/pressure	250 °C/65 bar
Flash separator#1 temperature/pressure	30.0 °C/64.4 bar
Flash separator#2 temperature/pressure	24.5 °C/0.2 bar
Crude methanol preheating	85 °C
Distillation column, number of stages	60
Distillation column, inlet pressure	1.3 bar
Compressors isentropic efficiency	75%
Pump efficiency	70%

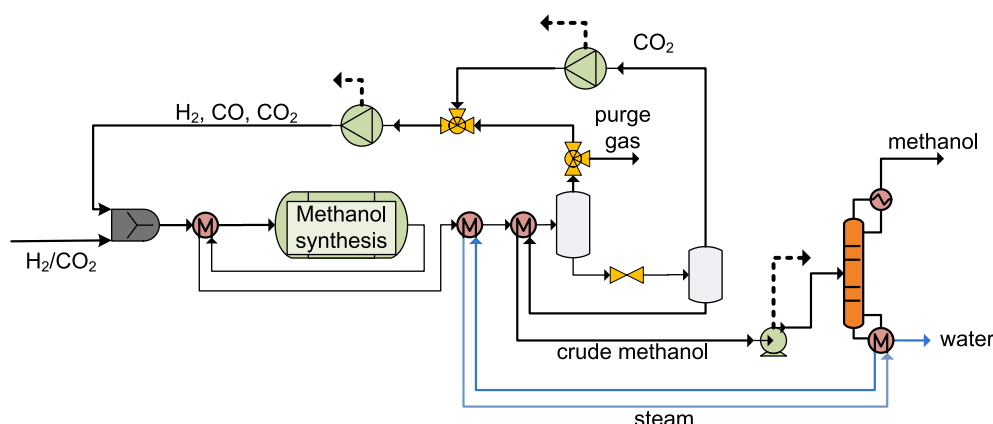


Fig. 2 – Process flowsheet of the CO₂ to MeOH concept.

reboiler demands are 397.4 kWh/tn_{MeOH}. This heat can be covered by gas cooling after the reactor and thus no additional heating system is required.

Regarding the economic evaluation, the methodology for the calculation of the equipment cost of each component in the CO₂ Utilization Unit (CUU) and the assumptions for Capex and Opex estimation are following. The Total Capital Investment (TCI) estimation is performed according to Peters & Timmerhaus methodology [32] and is based on a series of intermediate cost types, the first of which is the Total Purchased Equipment cost (TPEC) for each case. The equipment cost of each component is estimated based on similar equipment costs from the literature according to the following equation:

$$C_i = C_o \cdot \left(\frac{S_i}{S_o} \right)^f \quad (1)$$

Furthermore, having the Total Purchased Equipment Cost (TPEC) estimated, the next step is the Total Installed Cost (TIC) calculation. This is accomplished by multiplying each equipment cost with an installation factor n , specified for each component (see Table 3).

The parameters for the operational cost calculation such as O&M and insurance are considered as a portion of the Fixed Capital Investment (FCI). The assumptions that are made for the economic evaluation of the methanol plant are summarized in Table 4.

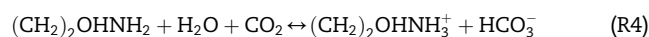
This (dimensionless) methodology for methanol plant cost estimation (both capital and operational) can be applied in large scale applications and it is adopted for the methanol

production cost. This approach is followed for the estimation of methanol plant costs at the examined scenarios below.

Pure CO₂ from coal derived flue gas

The Carbon Capture and Storage concept has attracted a lot of attention in the past decade. A great development has been gained on each of the main CO₂ separation techniques, even though no concept has reached to a commercial level yet. Among the several CO₂ caption options, amine scrubbing (MEA) and oxyfuel technology are considered the most competitive and ready to apply technologies for the first generation of applications in industrial scale. The post-combustion chemical absorption technique with amine scrubbing is the most mature technology with higher Technology Readiness Level and has already been tested and implemented in large scale applications [40]. Therefore, the amine scrubbing is selected as the CO₂ capture technology for this study.

The CO₂ chemical absorption from a flue gas is performed bringing a monoethanolamine (MEA) aqueous solvent into contact with CO₂ in a column, where the following reaction occurs:



The release of CO₂ carried in the solvent occurs in the stripper unit that operates at elevated temperatures, allowing the reverse reaction to take place. The MEA solvent is thus

Table 2 – Main characteristics of the methanol plant basic streams.

		Feed gas	Recycling gas	Reactor inlet	Reactor outlet	Methanol
Molar concentration	CO ₂	25.0%	15.3%	17.3%	13.5%	0.6%
	H ₂	75.0%	81.8%	80.3%	72.1%	0.0%
	H ₂ O	0.0%	0.1%	0.1%	6.1%	0.1%
	CO	0.0%	2.1%	1.7%	1.9%	0.0%
	CH ₃ OH	0.0%	0.7%	0.5%	6.5%	99.3%
Mole flow (kmol/kmol _{MeOH})		4.05	15.00	19.04	17.02	0.12
Pressure (bar)		65.3	65.3	65.1	65.0	1.1
Temperature (°C)		20.5	58.7	150.7	250.0	48.1

Table 3 – Equipment cost estimation for the methanol production unit.

Equipment list	Scaling parameter	Reference erected cost, C_o (M€)	Reference size S_o	Scale factor f	Installation factor n	Reference year	Ref.
Methanol reactor	Feed gas (kg/h)	7.69	87.5	0.6	2.1	2006	[28]
Heat exchanger	Heat duty (MW_{th})	39.26	355	1	1.49	2007	[33]
Distillation unit	Methanol flow rate (t/h)	16.58	6.75	0.7	1.52	2006	[28]
Compressor	Power (MW_e)	12.08	10	0.67	1.72	2006	[34]
Cooling system	Q rejected (MW_{th})	49.600	470	0.67	1.49	2007	[35]
Booster	Power (MW_e)	14.770	47.61	0.67	2.47	2011	[36,37]
Wastewater treatment	kg/hr water input	20.353	393,100	1.05	2.47	2010	[38]
Flash tank	Gas feed (kg/s)	Installation cost (in M€) = $2.47 \cdot 983.2 \cdot 10^{-6} \cdot (\text{feed gas in kg/s})^{0.8}$				2000	[39]

regenerated and reused. The required heat for the solvent regeneration is provided by steam extracted from the steam cycle. The gaseous outlet of the stripper mainly consists of CO_2 and steam, and these two components are separated by condensation at ambient temperature.

In this study, CO_2 stream is assumed that comes from a large scale CO_2 capture unit located in a coal fired power plant. The cost burden for CO_2 use as feedstock in a Power to Fuel concepts is included in the overall methanol production cost. Thus, the cost for CO_2 capture with the amine scrubbing post-combustion technology is assumed 43.8 €/tn $_{CO_2}$, which is the average CO_2 avoidance cost for coal-fired power plants, according to the widely accepted estimates performed by IEA [41]. This CO_2 cost includes the capture plant capital cost and the operational cost due to efficiency decrease of the main power plant. According to IEA's report the average post combustion capture technologies specific CO_2 capture capital cost is estimated at 19.5 €/tn $_{CO_2}$ and the corresponding specific operational cost is 24.3 €/tn $_{CO_2}$.

The delivered CO_2 stream bares the same specification with storage specification: oxygen should absent from the pure CO_2 stream to avoid ignition after reaction with the H_2 in the reactor. The specific heat demands for amine regeneration are 4.17 MJ $_{th}$ /kg $_{CO_2}$ and the electric power consumption 0.021 kWh/kg $_{CO_2}$ [42]. These heat and power consumptions are already included in the former analysis for the estimation of CO_2 avoidance cost and are not taken into account in the economic analysis in this study.

The second parameter that composes the cost of CO_2 is the transportation cost. The location of the CO_2 capture unit with respect to the CO_2 utilization unit influences the cost for CO_2 delivery. In case that the two units are in the same place, the cost of CO_2 equals to the cost for CO_2 separation (CO_2 delivery cost is zero). Otherwise, a specific cost 9.23 €/tn for CO_2 transportation is assumed based on [43] when a pipeline network is applied. From the same study [43], it is concluded that the specific cost for CO_2 storage in liquid form 4.46 to

13.86 €/tn CO_2 . Hence, in the present study, the total cost for CO_2 transport and constant when the CO_2 is conducted far away from the CO_2 capture plant is set 20 €/tn following the assumption made by Barbato et al. in Ref. [29].

Hydrogen production

In water electrolysis, water is split into O_2 and H_2 by means of electrical power. This option for H_2 production has the lowest efficiency (35–42%) and the highest H_2 production cost (for large scale 20–25 \$/GJ $_{H_2}$) among the other technological options. However, currently it is considered as the best option among the other RES based technologies for sustainable and clean hydrogen production in large scale [44,45]. There are three main electrolysis options (alkaline electrolysis, polymer electrolyte membranes and high-temperature electrolysis). Alkaline (KOH) electrolysis is the best available since it is quite mature up to large scale H_2 production [46]. Moreover, it has been tested successfully for discontinuous operation, and its load can be altered easily by adapting the current density [21]. Therefore, KOH electrolysis is selected as the most suitable technique for the examined concept in this study.

The water electrolysis process is not modeled in detail in this study. For the analysis, the basic aspects that affect the techno-economic evaluation are only taken into account: It is assumed that highly pure (>99.9%) H_2 and O_2 gaseous streams are produced in the electrolyzer, the operating conditions of which are 80 °C/30 bar, which are typical for large scale commercial alkaline electrolyzers [21,47,48]. The specific power consumption is set 4.34 kWh/Nm 3 based on [21]. Assuming that the rest of power demands in the electrolysis unit is 10% of the consumptions for water electrolysis and hydrogen production [47], the overall electrical consumptions are 55.56 kWh/kg $_{H_2}$.

Fig. 3 shows a schematic depiction of a H_2 production plant with water electrolysis. Apart from the electrolyzer which is the main component, the balance-of-plant also consists of several items: the rectifier, the control system, the water purification, and the water and alkaline tank. The process water is purified in the demineralizer before it enters the electrolyzer removing minerals and other impurities that may hurt the electrolyzer. As the electrolysis type that is selected is alkaline, a KOH tank is required in order to feed the electrolyzer with the electrolyte solution. The rectifier is served for the transformation of the alternating current (AC) to direct current (DC). The cooling water system is used for both the

Table 4 – Economic assumptions.

Exchange rate	0.755 €/€
Discount rate	10%
Recovery period	25 years
Capital Recovery Factor	0.11
Year basis	2012
Operating and Maintenance (O&M)	5% FCI
Insurance	2% FCI

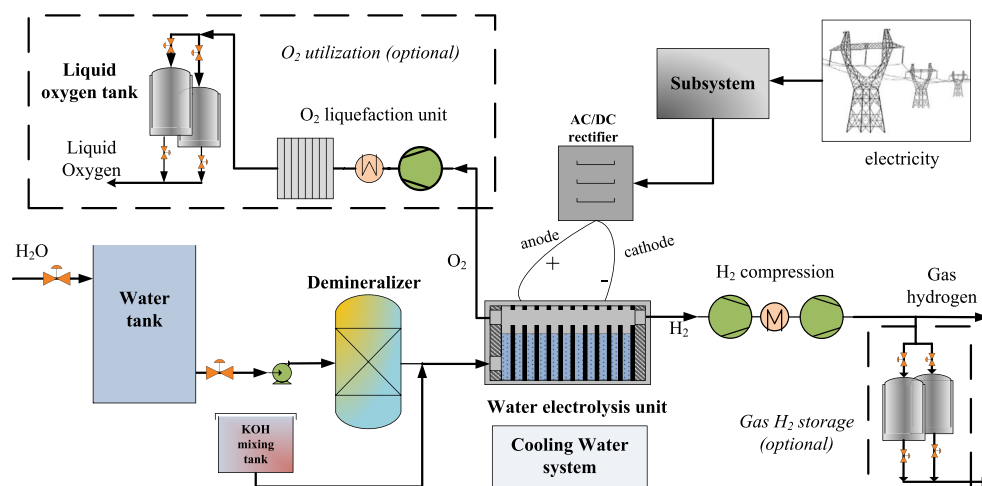


Fig. 3 – Conceptual process diagram of the water electrolysis plant.

maintenance of the electrolysis temperature at a constant level and the cooling of the product gas streams before compression and storage. Furthermore, the produced oxygen that is in high purity is considered as product and therefore, the oxygen liquefaction and storage is taken into account.

Investigation for the best hydrogen production scheme

Provided that water electrolysis is an energy demanding process, special effort is paid in this study in order to determine the scheme with the lowest specific hydrogen production cost. Table 5 summarizes the key parameters of a selected number of studies based on hydrogen production. The definition of the index capacity factor (C_f) is the fraction of the actual operation to the designed operation (i.e. if the electrolyzer operates at full load at 24/7 mode).

Table 5 illustrates the considerable discrepancy in the estimated hydrogen production costs even amongst studies with similar parameters. Furthermore special note has to be made to avoid wrong assumptions such as the use of renewable electricity from a “network” of wind farms in 24/7 mode or the use of a dam for hydrogen production with a capacity

factor more than 50–60%. In the following paragraphs, the aspects of some of the cases studies are discussed and evaluated in respect to the source of power for H_2 production.

Large scale electrolyzer with electricity from grid

This first scenario is based on central H_2 production with electricity from the grid feeding an electrolysis process (large scale) with a capacity of 140 MW_e or equivalent to the production of 60.5 t_{H_2}/day . Because of the size of the H_2 system, a central hydrogen production plant is considered with high capacity factor (90.0%). However, since the CCU plant is considered to have an 85% availability factor, the overall analysis is made with the latter value.

The methodology for the cost estimation of the electrolytic hydrogen production plant is adopted from the Production Case Studies carried out from the Department of Energy H2A Analysis and more specifically from the study on “Current Central Hydrogen Production from Grid Electrolysis version 3.0” [55]. According to this, the capital cost of a Central hydrogen production plant with similar design capacity to the case that is currently investigated (67,200 kg/d) was estimated \$69,775,218 or 784.0 €/kg/d (2010 basis).

Table 5 – Key parameters production costs of various hydrogen production plants.

Power source	Electricity price (€/kWh)	Capacity factor, C_f (%)	H_2 plant capacity	H_2 production cost (€/kg)	Year	Reference
Grid	0.004–0.047	98.6%	22 MW_e	2.29–3.75	1985	[49]
Grid	0.0543	90.0%	150,000 kg/day	4.113	2002	[48]
Grid	0.0560	100.0%	3.3 MW_e	3.270	2011	[21]
Grid	0.0427	85.0%	1500 kg/day	4.695	2005	[50]
Grid	0.0427	88.0%	1500 kg/day	3.989	2009	[47]
Grid	0.0233	70.0%	0.14 MW_e	1.994	1996	[51]
Grid	0.0475	97.0%	1500 kg/day	3.989	2013	[52]
Grid	0.0483	86.0%	50,000 kg/day	3.973	2013	[52]
RES	0.016	91.3%	100,500 MW_e	1.590	2003	[17]
Wind	0.0427	97.0%	>50,000 kg/day	3.492	2005	[50]
Wind	0.0349	98.0%	>50,000 kg/day	2.328	2009	[47]
Wind	0.0520	44.0%	50,000 kg/day	2.910	2011	[53]
Solar PV	0.047–0.064	27.1%	592,000 kg/day	4.291–5.028	2008	[54]
Hydropower	0.0104	87.7%	225,000 kg/day	1.558	2013	[29]

Table 6 – Regional difference in the electricity price and its impact on the H₂ economy.

	Unit	U.S.	E.U.	Japan
Electricity price (industrial 2013)	€/kWh	0.050	0.126	0.099
Electrolyzer cost	€/kg _{H2}	0.762	0.762	0.762
H ₂ compression & storage cost	€/kg _{H2}	0.043	0.043	0.043
Electricity cost	€/kg _{H2}	2.804	7.066	5.530
H ₂ production cost	€/kg _{H2}	3.609	7.871	6.334

Table 6 shows the different costs of a water electrolysis hydrogen production depending on the country. It is pointed out that the only varying parameter is the purchased electricity price from the national grid of each country.

The architecture of the total Power-to-Fuel scheme plays a very crucial role in the economic feasibility and implementation of the concept. In other words, the location of each major unit operation (H₂ plant, CO₂ capture and Carbon Utilization) is located determines the total cost.

According to several studies [48,56–58], the specific cost for hydrogen transport varies with the different scenario and the methodology for handling and transportation. For instance, transporting hydrogen in liquid form has very low cost (0.1–0.2 €/kg) but hydrogen liquefaction is a capital (800 €/kg/day) and energy (10–13 kWh/kg) intensive process [48]. Hence, it is suitable only for very long distance transmissions. On the other hand, the hydrogen handling in gaseous form has a specific capital cost equal to 2000 €/kW_e for compression with corresponding energy consumption 0.5–2 kWh/kg and 1.63 €/kg for transport via tube trailers and 0.2–0.6 Mio €/km via pipeline.

Table 7 presents the most suitable option for various cases of hydrogen transport, depending on size and transmission distance according to the study of Yang and Ogden [57]. It is concluded that for a large scale H₂ plant, the optimal way for H₂ transmission is through pipelines, regardless of the distance. Beyond this, according to the study of Tzimas et al. [56], the concept of centralized system that employs pipeline network for hydrogen delivery is expected to developed in the future.

For the evaluation of the whole “Power to Fuel” concept and the cost breakdown estimation, the determination of the location of the subsystems each other (CCU general configuration) needs to be defined. Table 8 presents various schemes with different configuration. The first three schemes are related to the main concept that is discussed in this section. It is concluded that, in terms of feedstock gas cost, hydrogen production scheme is preferable to be near to the unit that is consumed, i.e. the methanol plant. Thus, the first scenario is selected for the economic evaluation of the total concept with

the same main assumptions (energy source: grid electricity, electricity price: 0.05 €/kWh). As it is revealed from the methanol plant cost breakdown in Fig. 4, the hydrogen production has the lion's share on the methanol production cost. The terms and conditions for the minimization of the electrolyzer and electricity cost in order the technology to be more competitive is not part of this study.

Reflecting the contribution of CO₂ cost to the methanol production cost, the share of the CO₂ capture plant capital cost is 27.2 €/tn_{meOH} whereas the operation cost of the amine unit adds 34.0 €/tn_{meOH}. Concerning the methanol plant, the total annual operational cost are 3.5 M€/year whereas the total capital cost of the CCU plant (methanol production and recovery) is estimated at 82.38 M€ or 92.08€/tn_{meOH}. The most capital intensive equipment is the distillation unit (79.1% of the total equipment cost).

In the previous analysis, the market of oxygen is not taken into account. Oxygen is a valuable product that has numerous applications in various sectors such as the iron and steel industry, and the medical sector [59]. Given that oxygen that is produced through electrolysis is in high purity, it has the potential to fulfill the requirements for the implementation in every use (oxidizing agent or medical product). In this case, the compression, cooling and liquefaction of the produced oxygen and the corresponding power consumptions and operational costs are included. The specific power consumptions for O₂ liquefaction is set 0.3711 kWh/kg_{O2}. Because of the high quality in terms of purity, the oxygen selling price is set at 87.4 €/tn, based on [60]. The introduction of the oxygen liquefaction and compression unit causes 14.9% increase in the total capital cost and the corresponding total operational cost drops by 1.3% in the whole Power-to-Methanol system. As a consequence, the methanol production cost drops at 859.2 €/tn (10.1% decrease). It is seen that the oxygen marketing has a beneficial effect on the mitigation methanol production cost and should be taken into consideration.

Large scale electrolysis, plant using electricity from thermal power plant

Seeking for low electricity price for hydrogen production, a concept that has been recently emerged is the use of the “Power to Methanol” system as an energy storage option for the stabilization of the electricity system. In other words, in countries where the RES has a great share on the energy mix such in Germany, there is the need for the flexible operation of the thermal power plants, due to the stochastic behavior of the electricity production of the RES and the fluctuations in the energy demand. The conditions and under which the use of excess power can be profitable are examined in this section.

Table 7 – Comparative assessment for the best option for H₂ transport [57].

a/a	Size plant		Transmission distance		Preferable option for H ₂ delivery	Transmission cost estimation (€/kg)
	Small	Big	Short	Long		
1	X		X		gas tube trailer	0.60
2	X			X	liquid tanker truck	1.36
3		X	X		gas pipeline	0.23
4		X		X	gas pipeline	0.53

Table 8 – Summary of the several H₂/CO₂ production schemes (energy source: grid electricity, electricity price: 0.05 €/kWh).

Case	Hydrogen plant capacity	CCU configuration ^a	Inlet gas production & delivery cost (€/kg _{H₂-CO₂})
1	140 MW _e	H2P-CC-CUU in the same location	3.003
2	140 MW _e	H2P away from CC-CUU	3.337
3	140 MW _e	CC away from H2P-CUU	3.005
4	1 MW _e	H2P away from CC-CUU	3.982
5	9.4 MW _e	H2P away from CC-CUU	4.560

^a H2P: hydrogen plant, CC: CO₂ capture plant, CUU: CO₂ utilization unit.

The analysis is carried out based on data provided from the European Energy Exchange regarding the fluctuation of the electricity price in Germany in 2014 (see Fig. 5a) [61]. According to this, the average electricity price (AEP) for this year is 32.2 €/MWh. The economic evaluation is based on the term “clean dark spread”, (CDS), which is defined as:

$$CDS = AEP - COE [€/MWh] \quad (2)$$

where the Cost of Electricity (COE) is determined by the sum of the fuel cost, the cost from CO₂ emissions and other costs.

For simplicity reasons, several assumptions are made for the accomplishment of the economic evaluation of this case. The first is that the electricity pricing is the same for the whole examined period (see Fig. 5a) and is not affected by the introduction or not of the power plant to the energy system. The electricity cost and the earnings from its selling to the system are exclusively determined by the Equation (2).

The power plant that is used from this analysis is considered as a lignite fired thermal plant with a typical capacity of 300 MWe and a net efficiency of 36%, a typical value for German lignite power plants according to RWE [62]. As a starting point, the fuel cost is assumed as 2.0 €/GJ_{th}, the specific CO₂ emissions 0.971 tn_{CO2}/MWh, the CO₂ emissions tax 7.2 €/tn_{CO2} and the other operational costs are 2.0 €/MWh. Based on these assumptions, the COE is calculated at 29.0 €/MWh and hence, the average CDS = 3.3 €/MWh. As it seen in Fig. 5b, 27% of the year, the electricity production is not profitable to be sold to the energy system. Considering an availability factor of the plant 85% and making the arbitrary assumptions that it operates only at full load, the annual incomes from selling the electricity is 7.4 M€.

The H₂ plant capacity is the same with the power plant 300 MWe but the specifications for the economic evaluation the same with the previous scenario so that the comparison

between the two cases to be made on equal terms. At a first analysis, the H₂ plant operates for 2203 h, when the average COE is higher than the electricity price. Taking also into account the 85% capacity factor of the power plant, the corresponding capacity factor is calculated at 25.2%. The electrolyzer specific capital cost, according to the previous analysis, is 582.92 k€/MW_{th} and the electrolyzer efficiency is 50.0 kWh/kg_{H₂}. Similarly, the methanol unit has the same specific capital and operational cost with the previous case: 42.05 €/tn and 38.02 €/tn of methanol, respectively.

The decision for operating or not the “Power-to-Fuel” (PtF) system by the power plant operator is not taken suddenly or unexpectedly but it is programmed from the previous day, when the pricing of the electricity cost is determined by the electricity trading market. In this light and since the power to fuel plant operates in stable load, no storage units for H₂ and CO₂ are required.

The cost breakdown of the base case of this CCU concept that is studied is presented in Fig. 6. Contrary to the previous case where the electricity is supplied from the grid, the electrolyzer unit capital cost is the most significant factor for the determination of the minimum selling process of the

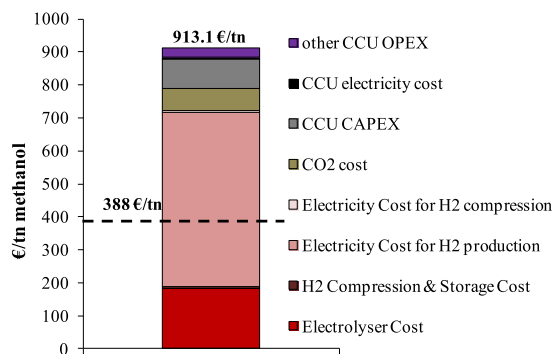


Fig. 4 – Power to Methanol cost breakdown.

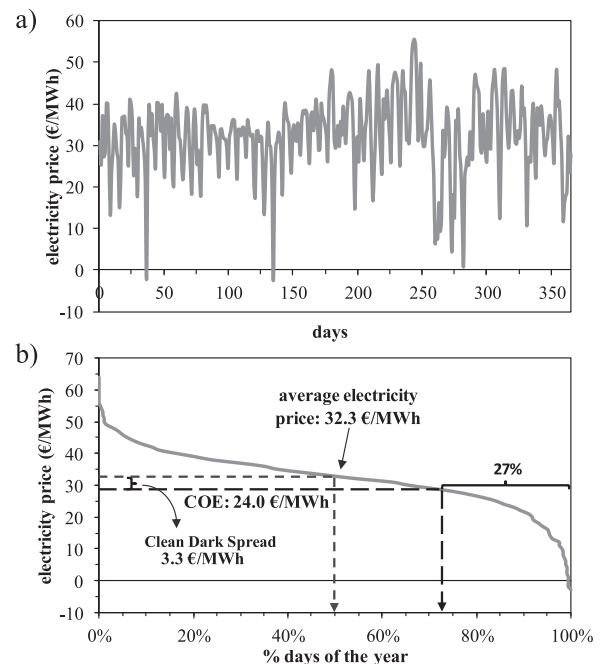


Fig. 5 – a) average daily electricity price and b) cumulated electricity price in 2014 in Germany.

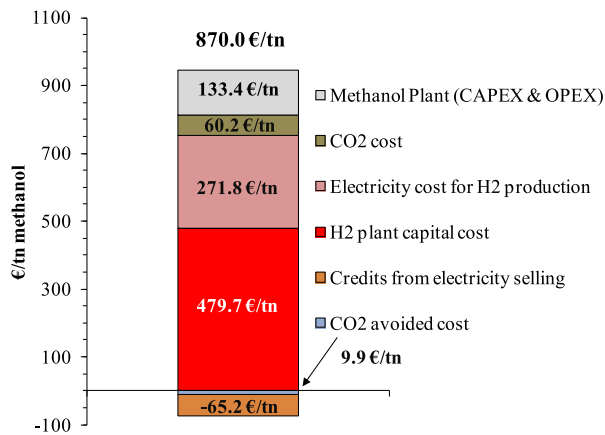


Fig. 6 – Methanol production cost break down (use excess electricity from lignite power plant).

methanol in this case. This is attributed to the lower hours that the PtF system operates and the electricity cost for hydrogen production is a bit lower.

A sensitivity analysis of the change of several important parameters by $\pm 20\%$ is performed. Fig. 7 presents how the minimum methanol selling price is affected. It is seen that the fuel cost and power plant performance have the greatest impact on the methanol price. Lower fuel price and power plant net efficiency increase the capacity factor of the PtF plant as the COE increases. Additionally, the reduction on the electrolyzer cost has a considerable effect on the methanol price minimization. Furthermore, the increase in CO₂ tax rate, which is expected the next years, beneficiates the feasibility of this concept.

It should be pointed out that in each case, only the under investigation parameter was changed, without affecting any other one. In most of these cases, this assumption is reasonable. For instance, changing the COE by changing the net efficiency or the fuel cost, the annual average electricity price is not expected to be affected in a large country with high national power load like Germany. On the other hand, any

change in the CO₂ emission credits will affect the electricity price, but this aspect is not easily determined because several factors related to energy policy and political issues should be taken into account, and therefore this correlation is neglected.

Decentralized H₂ production in small scale electrolysis plants using electricity from grid

In the case of a decentralized H₂ plants with smaller capacity than the corresponding in the previous case (central production), the hydrogen compression, storage and transport parts (C/S/T) should be taken into account. According to the discussion above, for small plant size and relative short distances between the H₂ plants and the central CUU, the most appropriate option for hydrogen handling and transport is the transport of compressed H₂ with gas tube trailers. In the case of providing electricity from grid, thanks to the 100% availability of power, the storage capacity is not large and therefore the installation cost of a H₂ storage unit is not taken into account. The capital cost of the unit was calculated based on the H2A analysis methodology for small forecourt plants with grid electrolysis in 2010 values.

Decentralized H₂ production in small scale electrolysis plants interconnected with RES (i.e. wind farm)

Contrary to the previous scenario, if electricity available is intermittent then a hydrogen buffering system to secure the continuous flow rate to the fuel synthesis plant is required. Hence, the H₂ compression storage and distribution components are also included in the investment cost analysis. In case that the H₂ plant is near to the CUU, the third part (i.e. hydrogen transport) is not taken into account.

For the investigation of this case, data of European Energy Exchange [61] obtained from Ref. [50] is used regarding a wind farm in Germany. This data is referred to the fluctuation of electricity price and the corresponding power feed-in per day (see Fig. 8a and b).

It should be underlined that, on the one hand, the electrolyzer should be highly flexible and have an effective operation at even far from the design point in order to be capable to catch up with the great fluctuations of the power output of the

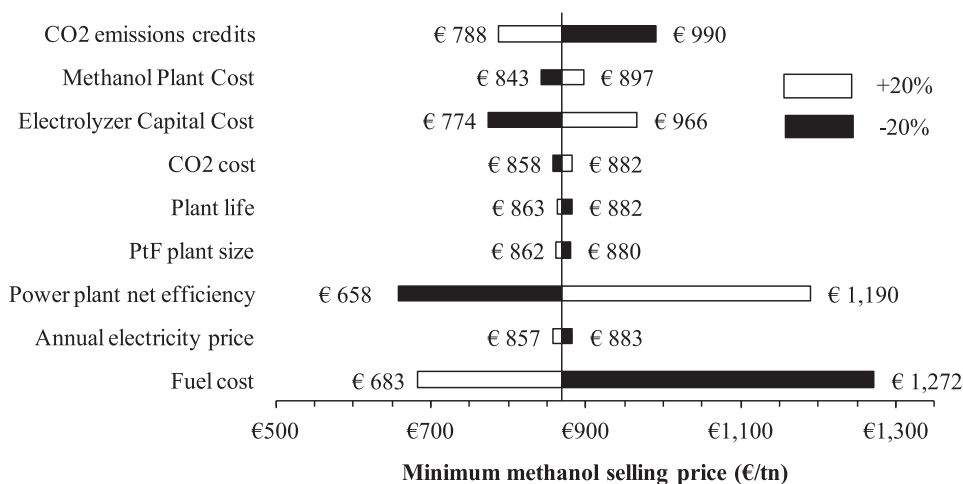


Fig. 7 – Using Power from Coal plant for Methanol Synthesis - Sensitivity analysis (the white bars are when an increase of 20% of each parameter is considered whereas the black bars for 20% decrease).

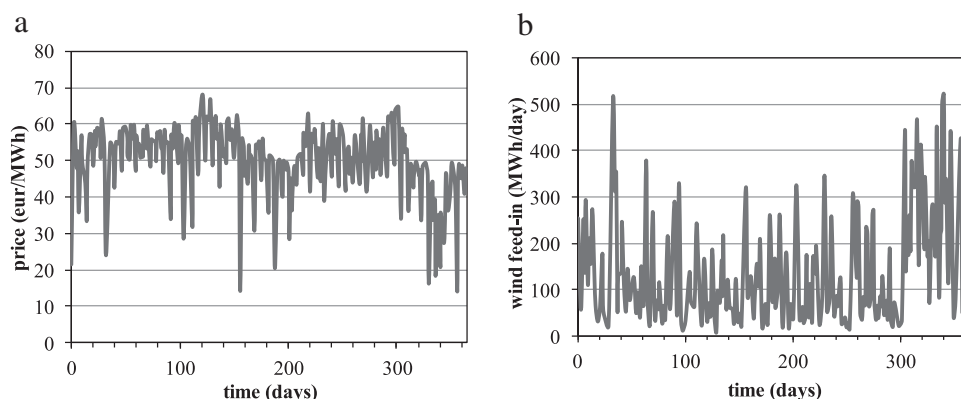


Fig. 8 – a Electricity price along one year period [50]. b Hourly wind in-feed in MW along one year period [50].

wind plant. On the other hand, the methanation unit is not technically proven that can be operate with flexibility under conditions with great variables in the feed rate, since it consists of several equipment and complex heating & cooling network systems that perform effectively mainly in the design point. Hence, the hydrogen storage is necessary in order to secure the H_2 feeding with constant flow rate.

The demand for a constant hydrogen flow rate to the CO_2 Utilization Unit is the main input for the determination of the hydrogen storage capacity. For each case, the hydrogen output flow rate is the mean value of the annual hydrogen production. On the other hand the tank should be big enough in order to can host all the quantity of the produced hydrogen anytime. The hydrogen production curve is calculated based on the data from only one year presented in Fig. 8a and b, assuming to be representative for the whole plant life. The specific cost for the H_2 storage is set at 400 €/kg [48].

Fig. 9a shows how much electrolyzer operation time is required for different plant sizes versus the maximum price allowed for the electricity purchased. It is clear that by producing H_2 only when the electricity price is low leads to corresponding low annual hydrogen production (i.e. small capacity factor of the electrolysis plant). This negatively

affects the payback capability because high capacity electrolyzers (with high capital cost) remain idle for prolonged fraction of the annual time (Fig. 9b).

Fig. 10 summarizes the H_2 production cost breakdown for electrolyzers with different sizes. It is clear that the cost for hydrogen storage plays the most significant role in the hydrogen production cost when the constant hydrogen flow rate is considered as a necessary condition. Moreover, it is proven that the maximization capacity factor of the unit, which is achieved with small size units, is a more important parameter compared to the maximization of the utilization of the available electricity produced at the RES plant.

Concluding remarks

Each of the three main parameters for the determination of the H_2 cost (electrolyzer capital cost, electricity cost and storage cost) play a key role in the feasibility of the scheme. When the capacity factor of the hydrogen plant is maximum (supply of grid electricity in large scale application) the minimum methanol cost is determined by the electricity cost. When cheap electricity is used in large scale application but

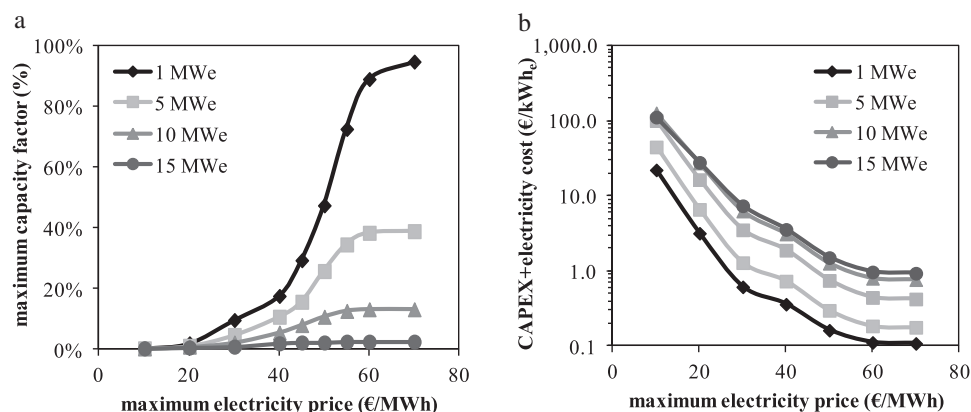


Fig. 9 – a Dependence of the maximum capacity factor of the electrolyzer on the maximum purchased electricity price. b Dependence of the capital cost and electricity cost with the maximum purchased electricity price.

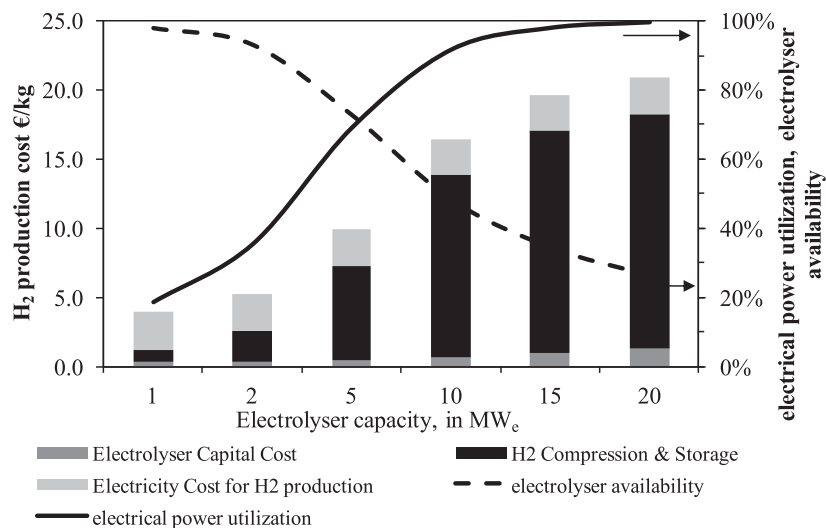


Fig. 10 – Sensitivity analysis on Electrolysis plant capacity (decentralized H₂ plant using electricity from RES).

with less operating time, the electrolyzer capital cost is the most important factor. Finally, when RES electricity is used in an autonomous H₂ plant, away from the methanol plant, the dominant factor is the H₂ management in terms of storage issue. The concept of central electrolysis using grid electricity preferably in the same site with the CO₂ capture and utilization units is the most appropriate in terms of H₂ production cost and consequently, the methanol production cost.

Fig. 11 illustrates the impact of the electricity price and the capacity factor (C_f) on the feasibility prospects of a Power to Methanol concept. Additionally, based on the typical ranges of cost of electricity and the operating hours for various power sectors [63,64], the impact of the electricity provider to the hydrogen plant on the methanol production cost is also depicted. The maximization of the plant operation hours is of

the most important requirements, since in case of low capacity factors, the methanol production cost becomes extremely high even in the case of low expensive or zero electricity cost. For this reason, concepts based on renewable hydrogen with high electricity cost and low capacity factor are not economically viable and thus are not suggested for the time being. It is concluded that the use of cheap electricity derived from thermal plants (coal or nuclear) under a regular basis has the potential to produce methanol from CO₂ with high prospects of competitiveness. Similarly, in case that electricity is provided from the grid, the region in which the concept is examined plays considerable role, because electricity pricing is a matter of policy of each state/country.

The current status of this concept is not economically attractive under any conditions studied in here. The production

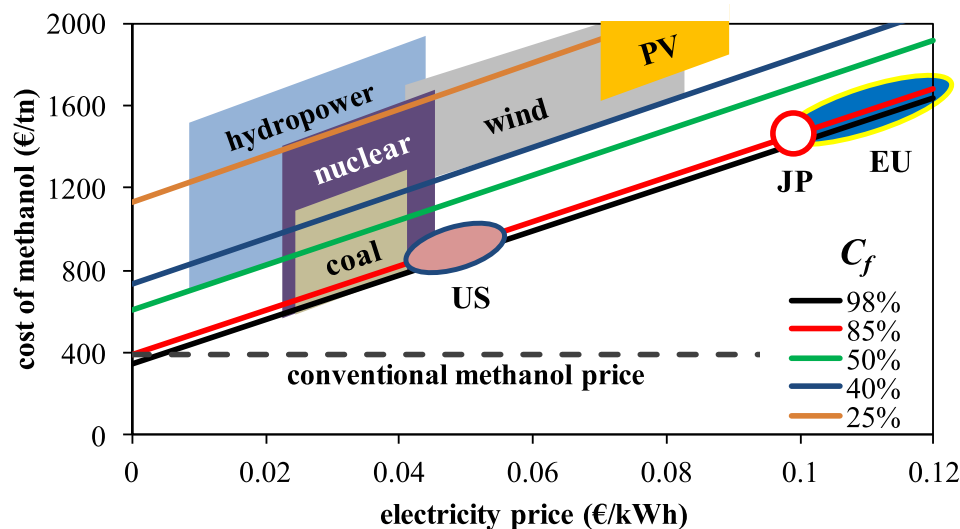


Fig. 11 – Influence of electricity price and capacity factor (C_f) on the methanol production cost. (US, JP, EU stands for the electricity grid provided in USA, Japan and EU-28, respectively).

cost of methanol from CO₂ is about 2.5 times higher than the corresponding conventional prices. A considerable effort is needed in order the CO₂ derived fuels to reach a competitive level in the global market. This study aimed to define the barriers that should overcome in order to become feasible providing charting of electricity source per case, charting of gas storage and Capex/Opex estimation and cost breakdown. More specifically, the key parameters that strongly affect the viability of this CO₂ utilization concept are the following:

- **Electricity cost.** The electrical consumptions that are required in order to produce hydrogen from water and then to transform CO₂ into organic compounds with high energy content is the most significant factor that influences the power to fuel economic feasibility. In terms of efficiency enhancement, the state-of-the art electrolysis reaches the maximum achievable value (~70%), which is thermodynamically limited. Consequently, only the use of cheap electricity on regular basis (high capacity factor) can contribute to the minimization of the share of the electricity cost on the methanol/ethanol production cost.
- **System configuration.** It is revealed in this study that the hydrogen production is preferable to be near to the methanol plant, avoiding high logistics costs for delivery and storage of H₂. On the other hand, the location of the CO₂ capture unit has small influence on the inlet gas production and delivery cost.
- **Oxygen utilization.** The oxygen that is produced in the H₂ plant is in high quality and can be used in numerous applications in industrial, medical and other sectors. Selling the produced O₂ with a very competitive price, the methanol production cost reduces at least 10%.
- **Capex reduction.** The investment cost has a 30% share on the methanol production cost. The 2/3 of this is the alkaline electrolyzer cost. So, a new electrolyzer design with lower cost would reduce considerably the minimum selling price of the produced alcohol in order the investment to be profitable. A revolutionary and alternative route of CO₂ reduction to value added chemical synthesis is the technology based on microbial electrosynthesis system (MES) where microorganisms are used in order to convert the direct electric current into chemical energy such as ethanol, directly from water and CO₂. On the one hand, this route is expected to have reduced investment cost as the water-to-ethanol process is conducted in only one step. On the other hand, since this is a relative new concept, it has not been implemented in large scale yet so that a complete economic assessment and comparison to be accomplished.

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